ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO1 ENGINEERING DESIGN ASSIGNMENT

Standards-Compliant Optical Fiber Link Design: From Communication Requirement to Deployment Decision

60 km · 1550 nm · 10 Gb/s with 100 Gb/s upgrade provision

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Course Outcome CO1 · Open-Ended Engineering Design Assignment · Maximum Marks: 100

60 km	1550 nm	10 Gb/s	100 Gb/s
Link length	Wavelength	Data rate	Upgrade ready

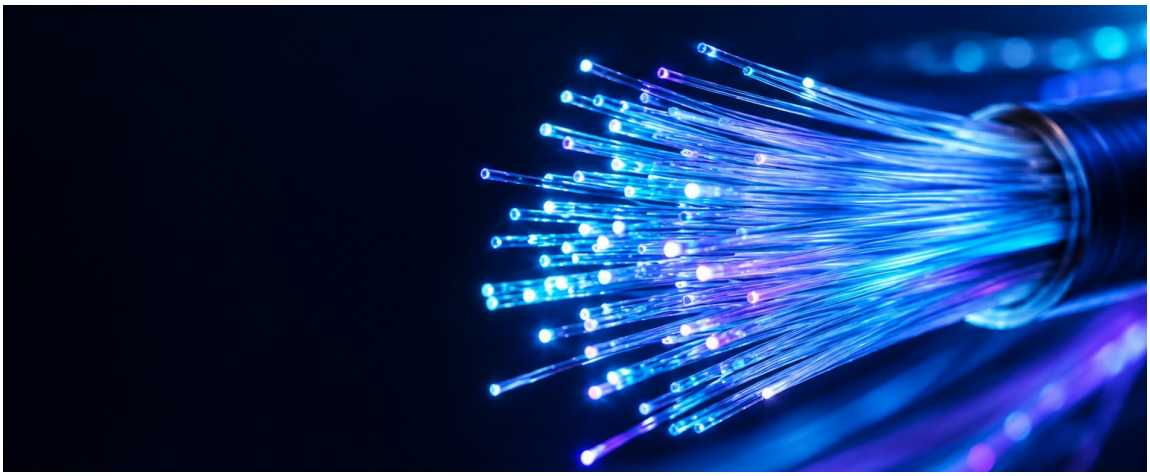


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1. INTRODUCTION

Optical fiber communication carries information as pulses of light guided inside a thin glass fiber. Light stays inside the central **core** because the core has a slightly higher refractive index (n_1) than the surrounding **cladding** (n_2), so rays hitting the boundary at shallow angles undergo total internal reflection. Compared with copper, fiber offers very low loss, very large bandwidth and immunity to electromagnetic interference, which is why it is used for telecom backbones, access networks and business links.

Why 1550 nm? Silica fiber has its minimum attenuation near 1550 nm, where the two intrinsic loss mechanisms (Rayleigh scattering falling with wavelength, infrared absorption rising with wavelength) balance. The same window is where erbium-doped fiber amplifiers operate, so a 1550 nm link can be amplified without converting the signal back to electrical form. The price paid is chromatic dispersion: standard G.652.D fiber has D = 17 ps/(nm·km) at 1550 nm, which is the main problem this design must solve.

The requirement. A service provider must connect a central office to a business building by a **60 km** link at **1550 nm** carrying **10 Gb/s** (bit period T_b = 100 ps), with a provision to upgrade to **100 Gb/s** later (T_b ≈ 10 ps). The fiber must be a standard ITU-T category, tolerate tight bends at the building entry, and keep pulse integrity, polarization performance and signal quality (OSNR).

Why design and verification are needed. A link that "looks fine" on one parameter can still fail on another. For example, the 60 km power budget alone is comfortable, but uncompensated dispersion smears every 100 ps bit across about four neighboring bit slots, and the dispersion-compensating fiber that cures this then eats most of the power margin. Only a complete chain of calculations (modal condition → power budget → dispersion/PMD → OSNR → verification) can show whether the link is deployable. This report follows that chain, as required by the assignment:

REQUIREMENT → ALTERNATIVES → ANALYSIS → SELECTION → LINK DESIGN → VERIFICATION → DEPLOYMENT DECISION



Figure 14. Illustrative field-level fiber handling and termination work (AI-generated visual; not a record of this proposed installation).

2. ENGINEERING DESIGN PROBLEM

2.1 Problem statement (as given in the assignment)

THE ENGINEERING DESIGN PROBLEM

A communication service provider plans to establish a 60 km optical fiber link operating at 1550 nm to carry 10 Gb/s traffic between a central office and a business building, with provision for a later upgrade to 100 Gb/s. The link must use standard ITU-T fiber categories, tolerate tight bends at the building entry, and maintain pulse integrity, polarization performance and signal quality. The design engineer must determine whether the proposed link configuration is technically suitable before deployment.

The assignment states that the following are **not given** and must be decided by the designer: (i) the final fiber type, ITU-T category and propagation choice; (ii) the mode-selection decision; (iii) the dominant loss mechanism requiring attention; (iv) whether dispersion compensation is required, its length, and whether PMD is acceptable; (v) whether the proposed amplification strategy is acceptable from an OSNR perspective; and (vi) the final deployment recommendation. These are answered through Design Decisions 1–4 and the Final Engineering Decision.

2.2 Marks distribution

Table 1: Marks distribution (rubric of the assignment)

Stage / criterion	Marks	Bloom level	Report section
Stage 1 – Interpret the communication requirement	10	K3/K4	Section 3
Stage 2 – Fiber / propagation selection	15	K4/K5	Section 4
Stage 3 – Optical power budget	15	K4/K5	Section 5
Stage 4 – Dispersion / PMD management	15	K4/K5	Section 6
Stage 5 – OSNR and amplification	10	K5	Section 7
Final system design – create the integrated link	20	K6	Section 9
Stage 6 verification and final engineering decision	15 (5 + 10)	K5/K6	Sections 8 and 10
Total	**100**		

2.3 Given parameters

Table 2: Complete list of given parameters

Parameter	Given value
Transmission distance	L = 60 km
Operating wavelength	λ = 1550 nm
Data rate / bit period	10 Gb/s, T_b = 100 ps (upgrade to 100 Gb/s, T_b ≈ 10 ps)
Candidate fiber A (step-index)	n_1 = 1.4500, n_2 = 1.4450, core radius a = 4.0 µm
Candidate fiber B (step-index)	n_1 = 1.480, n_2 = 1.465, core radius a = 25 µm
Absorption loss	0.04 dB/km
Rayleigh scattering loss	0.11 dB/km
Bending loss (distributed)	0.03 dB/km
Transmitter power	+3 dBm
Connectors	Four connectors, 0.4 dB each
Splices	Ten splices, 0.08 dB each
Building-entry bend loss	0.8 dB with G.652.D-type fiber; 0.1 dB with G.657.A-type fiber

Receiver requirement	Sensitivity = −20 dBm; required system margin = 6 dB
Chromatic dispersion	G.652.D: D = 17 ps/(nm·km); G.655: D = 4.5 ps/(nm·km)
Source spectral width	Δλ = 0.4 nm
Input pulse width	30 ps
Dispersion-compensating fiber (DCF)	D_DCF = −100 ps/(nm·km); attenuation = 0.5 dB/km
PMD	PMD coefficient = 0.1 ps/√km; PMD limit = 0.1·T_b
Birefringence	B = 1×10^{-7} (standard fiber); B = 4×10^{-4} (PM fiber); beat length L_B = λ/B (see note)
OSNR study (0.1 nm reference bandwidth)	Signal power = −16 dBm; ASE noise power = −39 dBm; required OSNR = 17 dB
Optical amplifier	Signal gain = 14 dB; ASE noise increase = 16 dB

Note on the birefringence exponents
The exponents of B are not legible in the supplied assignment PDF (the text reads “B = 1 × 10” and “B = 4 × 10”). The standard textbook values **B = 1×10^{-7}** (standard fiber) and **B = 4×10^{-4}** (PM fiber) are therefore adopted in this report. Should the assignment specify different exponents, only Section 6.H (the beat lengths) would change; no other result is affected.

2.4 Design workflow

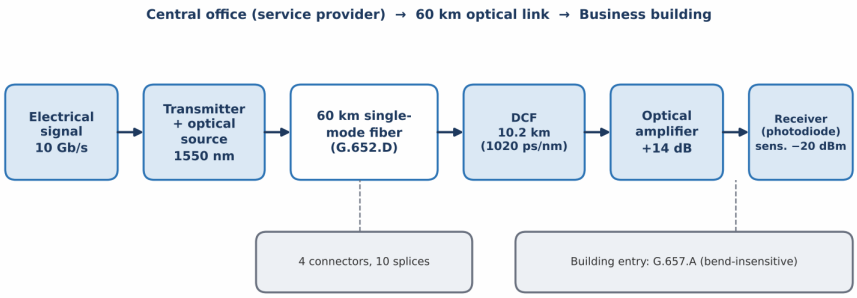


Figure 1: Overall 60 km optical-fiber communication system block diagram (central office to business building).

The signal chain examined in this report is shown in Figure 2 with the optical power level at each point; the individual values are derived in Sections 5 to 7.

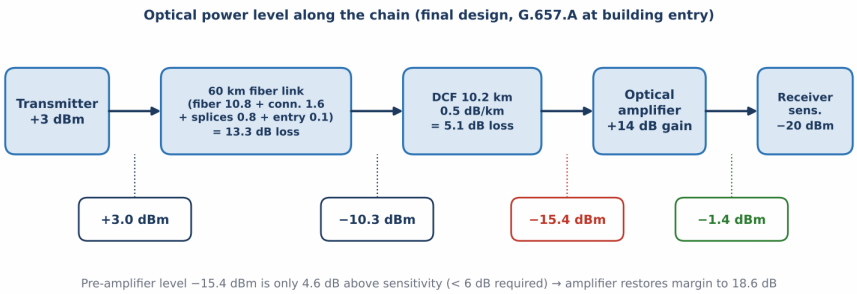


Figure 2: Transmitter → optical fiber → dispersion compensation → amplifier → receiver architecture with the optical power level at each point (values derived in Sections 5–7).

3. STAGE 1 – INTERPRET THE COMMUNICATION REQUIREMENT

3.1 Optical-fiber fundamentals

Table 3: Optical-fiber fundamentals

Concept	Meaning	Why it matters for this link
Ray nature	Light modeled as straight rays reflecting at the core–cladding boundary (total internal reflection). Valid when the core is much larger than the wavelength.	Explains acceptance angle, NA and why a multimode core (Fiber B) has many different path lengths.
Wave nature	Light is an electromagnetic wave; the fiber supports only discrete field patterns (modes) found from Maxwell's equations. Needed when the core is comparable to λ.	Fiber A (a = 4 µm, λ = 1.55 µm) must be analysed as a wave guide; only the LP_{01} mode survives.
Coherence	A laser has a narrow spectral width (long coherence); an LED is broad.	Δλ = 0.4 nm sets the chromatic-dispersion broadening (Δτ = D·Δλ·L).
Polarization	Orientation of the electric field. Two orthogonal polarization states propagate in a single-mode fiber.	Small birefringence makes them travel at slightly different speeds → PMD.
Numerical aperture (NA)	NA = $\sqrt{(n_1^2 - n_2^2)}$ = sin θ_max; measures light-acceptance.	Larger NA collects more source power but supports more modes.
V-number	V = (2πa/λ)·NA; normalized frequency combining core size, wavelength and index difference.	V < 2.405 single-mode.
LP modes	Linearly polarized mode groups (LP_{01}, LP_{11}, …) that approximate the exact HE/EH/TE/TM modes in weakly guiding fiber ($n_1 \approx n_2$).	Number of LP modes grows with V; each carries energy at its own speed.
Cut-off	Each mode exists only above its cut-off V. The first higher mode (LP_{11}) cuts off at V = 2.405 (first zero of J_0).	Defines the single-mode region and the cut-off wavelength λc = 2πa·NA/2.405.
Propagation characteristics	Attenuation, dispersion (modal, chromatic, polarization-mode) and nonlinearity along the guided path.	Together they set the maximum distance × bit-rate of the link.

3.2 Fiber standards (ITU-T)

Table 4: ITU-T fiber categories relevant to the project

Standard	Fiber type	Key features
ITU-T G.652.D	Standard single-mode fiber (non-dispersion-shifted), low water peak	Zero dispersion near 1310 nm; D ≈ 17 ps/(nm·km) at 1550 nm; cable cut-off wavelength ≤ 1260 nm; the most widely installed access/metro/long-haul fiber; low PMD coefficient specified.
ITU-T G.655	Non-zero dispersion-shifted fiber (NZDSF)	Small but non-zero dispersion in the C-band (given here: 4.5 ps/(nm·km)); designed to reduce four-wave mixing in DWDM systems and to need less compensation.
ITU-T G.657	Bend-insensitive single-mode fiber (categories A1, A2, A3 and B2, B3)	Much lower bending loss at small radii (about 10 mm for A1, 7.5 mm for A2, 5 mm for B3); the A categories are compatible with G.652.D and can be spliced to it; intended for access networks and building entry.

3.3 Transmission impairments

Table 5: Impairments, effects and control methods

Impairment	Cause	Effect on the link	Control

Absorption loss	Silica infrared tail, OH^- ions and impurities	Power loss, 0.04 dB/km	Pure silica, low-water-peak fiber
Rayleigh scattering	Microscopic density fluctuations; $1/\lambda^4$	Power loss, 0.11 dB/km (largest intrinsic term)	Operate at long wavelength (1550 nm)
Bending loss	Macrobends/microbends push light out of the core	0.03 dB/km distributed + building-entry bend loss	Bend-insensitive G.657, bend-radius control
Chromatic dispersion	Different wavelengths (source Δλ) travel at different speeds	Pulse broadening $\Delta\tau = D \cdot \Delta\lambda \cdot L \rightarrow$ ISI	Narrow source, G.655, DCF
Modal dispersion	Different modes have different path lengths	Strong pulse broadening in multimode fiber	Single-mode fiber (or graded-index)
PMD	Birefringence splits two polarization modes	Random differential delay √L	Low-PMD fiber, PMD compensation, coherent DSP
OSNR	Amplifier ASE noise added to signal, measured in 0.1 nm	Low OSNR → low Q-factor → high BER	Adequate launch power, low-noise amplifier, gain placement
BER	Probability that a received bit is wrong	Must stay below the design target (e.g. 10^{-9}–10^{-12})	Sufficient power margin, OSNR and pulse integrity

3.4 Practical constraints

- **Bend radius:** standard G.652.D fiber is normally specified at about 30 mm radius; a building entry often needs sharper bends (this is why the assignment gives 0.8 dB for G.652.D but only 0.1 dB for G.657.A).
- **Upgrade path:** at 100 Gb/s the bit period is only ≈ 10 ps, so the PMD limit (0.1·T_b) tightens from 10 ps to 1 ps, dispersion tolerance becomes ten times smaller and a higher OSNR is needed.
- **System margin:** the received power must exceed the receiver sensitivity (−20 dBm) by at least 6 dB to cover aging, repairs, temperature and connector re-mating.
- **Standards compliance:** every fiber used must belong to an ITU-T category so that components are interoperable and available.

DESIGN REQUIREMENT STATEMENT

The proposed 60 km, 1550 nm link must (1) use a **single-mode, ITU-T standard fiber** ($V < 2.405$) so that modal dispersion is absent; (2) deliver at least **−14 dBm** at the receiver (sensitivity −20 dBm plus 6 dB margin) after all fiber, connector, splice and building-entry losses; (3) keep the broadened pulse **below the 100 ps bit period** (net chromatic dispersion compensated to first order); (4) keep **PMD ≤ 0.1·T_b** (10 ps now, 1 ps for 100 Gb/s); (5) keep **OSNR ≥ 17 dB** after amplification; and (6) use **bend-insensitive G.657.A fiber at the tight building entry** while leaving an upgrade path to 100 Gb/s.

Viva tip

If asked “why 1550 nm?” answer: lowest fiber loss (Rayleigh falling, absorption not yet rising) and the erbium amplifier window; its drawback is higher chromatic dispersion in G.652.D.

4. STAGE 2 – FIBER / PROPAGATION SELECTION

A. Numerical aperture

A.1 NA of Fiber A

Given	$n_1 = 1.4500$, $n_2 = 1.4450$, $a = 4.0\ \mu m$
Formula	$NA = \sqrt{(n_1^2 - n_2^2)}$
Substitution	$NA = \sqrt{(1.4500^2 - 1.4450^2)} = \sqrt{(2.102500 - 2.088025)}$
Calculation	$NA = \sqrt{0.014475}$ Acceptance half-angle in air: $\theta_{max} = \sin^{-1}(NA) = \sin^{-1}(0.1203)$
Answer	**NA(A) = 0.1203, $\theta_{max} \approx 6.91°$**
Engineering Interpretation	Fiber A accepts light only within a narrow cone (≈ ±6.9°). This is typical of single-mode fiber: light must be launched carefully (laser source, good alignment), but the small index step (Δn = 0.005) keeps the guided field wide enough for a 4 µm core to be single-mode.

A.2 NA of Fiber B

Given	$n_1 = 1.480$, $n_2 = 1.465$, $a = 25\ \mu m$
Formula	$NA = \sqrt{(n_1^2 - n_2^2)}$
Substitution	$NA = \sqrt{(1.480^2 - 1.465^2)} = \sqrt{(2.190400 - 2.146225)}$
Calculation	$NA = \sqrt{0.044175}$ $\theta_{max} = \sin^{-1}(0.2102)$
Answer	**NA(B) = 0.2102, $\theta_{max} \approx 12.13°$**
Engineering Interpretation	Fiber B has a much larger NA and a larger core, so it couples light from LEDs or lasers easily — but a large NA together with a large core means many modes (next sections).

B. V-number at 1550 nm

B.1 V-number of Fiber A

Given	$a = 4.0\ \mu m$, NA = 0.1203, $\lambda = 1.55\ \mu m$
Formula	$V = (2\pi \cdot a / \lambda) \cdot NA$
Substitution	$V = (2\pi \times 4.0\ \mu m / 1.55\ \mu m) \times 0.1203$
Calculation	$2\pi \times 4.0 = 25.133\ \mu m$; $25.133 / 1.55 = 16.215$; $16.215 \times 0.1203 = 1.951$
Answer	**V(A) = 1.951 (dimensionless)**
Engineering Interpretation	V(A) is below 2.405, so only the fundamental LP_{01} mode (two polarizations) is guided. The pulse is carried by one mode only.

B.2 V-number of Fiber B

Given	$a = 25\ \mu m$, NA = 0.2102, $\lambda = 1.55\ \mu m$
Formula	$V = (2\pi \cdot a / \lambda) \cdot NA$
Substitution	$V = (2\pi \times 25\ \mu m / 1.55\ \mu m) \times 0.2102$
Calculation	$2\pi \times 25 = 157.08\ \mu m$; $157.08 / 1.55 = 101.34$; $101.34 \times 0.2102 = 21.30$
Answer	**V(B) = 21.30**

Engineering Interpretation	V(B) is almost nine times larger than the single-mode limit. The fiber is strongly multimode: a large core and a large NA support hundreds of modes.

C. Mode analysis (single-mode or multimode?)

Table 6: Modal classification

Fiber	V at 1550 nm	Condition	Result
A	1.951	V < 2.405	**Single-mode** (only LP_{01})
B	21.30	V 2.405	**Multimode** (many LP modes)

The single-mode condition comes from the zero of the Bessel function J_0: the first higher-order mode (LP_{11}) is cut off at V = 2.405. Fiber A is below this value, Fiber B far above it.

D. Mode count of Fiber B

D.1 Approximate number of guided modes (step-index)

Given	V(B) = 21.30; step-index profile
Formula	$M \approx V^2 / 2$ (step-index); $M \approx V^2 / 4$ for a graded-index (parabolic) profile
Substitution	$M \approx (21.30)^2 / 2 = 453.7 / 2$
Calculation	For comparison (graded-index): $M \approx 453.7 / 4 = 113$
Answer	**M ≈ 227 guided modes (step-index Fiber B)**
Engineering Interpretation	Each of the ≈ 227 modes travels along a different effective path, so a single input pulse is split into 227 slightly delayed copies. Section 6.A shows that this limits Fiber B to a few hundred kb/s over 60 km, so it is unsuitable for 10 Gb/s.

E. Cut-off condition and cut-off wavelength of Fiber A

Single-mode operation requires **V < 2.405**. Since $V = 2\pi a \cdot NA/\lambda$, the condition V = 2.405 gives the **cut-off wavelength** λc above which only LP_{01} propagates: $\lambda c = 2\pi a \cdot NA / 2.405$.

E.1 Cut-off wavelength of Fiber A

Given	a = 4.0 µm, NA = 0.1203, V_c = 2.405
Formula	From $V = 2\pi a \cdot NA/\lambda$ at V = 2.405: $\lambda c = 2\pi a \cdot NA / 2.405$
Substitution	$\lambda c = (2\pi \times 4.0\ \mu m \times 0.1203) / 2.405$
Calculation	Numerator = 25.133 × 0.1203 = 3.0237 µm λc = 3.0237 / 2.405 = 1.2573 µm
Answer	**λc ≈ 1.257 µm = 1257 nm**
Engineering Interpretation	The operating wavelength 1550 nm is above λc (1550 nm > 1257 nm), so Fiber A is single-mode at 1550 nm. The value also lies just under 1260 nm, the maximum cable cut-off wavelength specified for ITU-T G.652 fibers, which is consistent with Fiber A being a G.652.D-type fiber. (For Fiber B, the same formula gives 13.7 µm, which is why it is multimode at 1.55 µm.)

F. Ray theory versus wave optics

Table 7: Ray theory versus wave optics

Model	Basis	Useful when	Limitation

Ray (geometrical) optics	Straight rays, Snell's law, total internal reflection, NA, acceptance angle	Core λ and V large (many modes) — Fiber B (V = 21.3)	Cannot describe single-mode fields, mode cut-off, mode-field diameter or polarization effects
Wave optics	Maxwell's equations → wave equation → guided modes (LP modes), V-number, cut-off	Core comparable to λ and V small — Fiber A (V = 1.95)	More mathematics; for many modes the results approach the ray picture

Because V(B) = 21.3 gives hundreds of modes, ray theory is a good approximation for Fiber B. For Fiber A (V = 1.95, only one mode, core diameter 8 µm ≈ 5λ) ray theory fails and the wave (LP-mode) model is required. The **selected fiber is single-mode, so the wave-optics model is the appropriate one.**

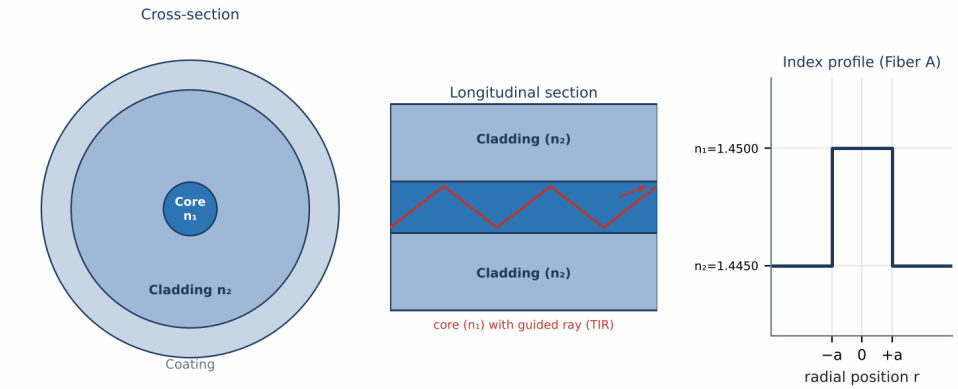


Figure 3: Step-index fiber structure: cross-section, longitudinal section with a guided ray, and refractive-index profile (Fiber A values).

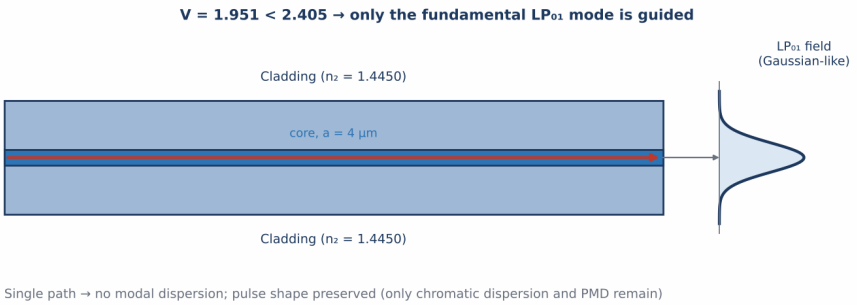


Figure 4: Single-mode fiber propagation: one guided mode (LP_{01}), V = 1.951.

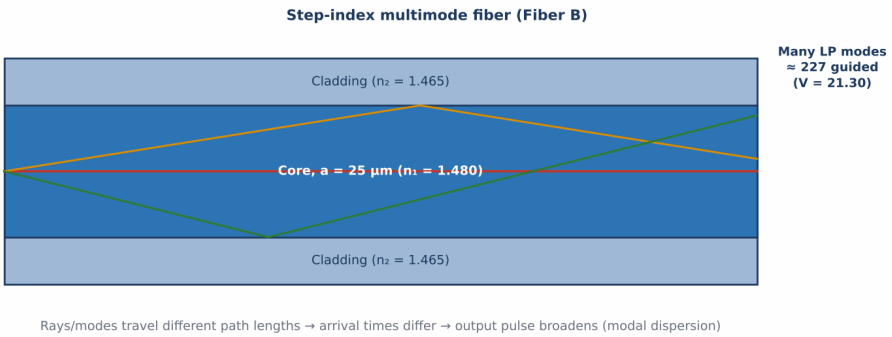


Figure 5: Multimode step-index fiber propagation: many rays/modes with different path lengths (Fiber B, V = 21.30).

G. Fiber comparison

Table 8: Comparison of propagation types

Characteristic	Single-mode (SMF)	Step-index multimode	Graded-index multimode
Core characteristics	Small core (≈ 8–10 µm); here a = 4 µm; Δn small	Large core (50–62.5 µm typ.); here a = 25 µm; uniform n_1	Large core; refractive index decreases parabolically from center to cladding

Mode behaviour	One mode (V < 2.405)	Hundreds of modes (≈ $V^2/2$ = 227)	Many modes (≈ $V^2/4$ = 113 for the same V)
Modal dispersion	None	Very large: Δτ ≈ 3.03 µs over 60 km (Fiber B)	Reduced: Δτ ≈ 3.80 ns over 60 km (Fiber B parameters)
Bandwidth–distance	Highest (limited by chromatic dispersion and PMD)	Lowest (≈ 165 kb/s over 60 km)	Moderate (≈ 130 Mb/s over 60 km)
Source coupling	Laser, careful alignment	Easy (LED or laser)	Easy
Typical application	Telecom, long-distance, access, metro	Short links, low-speed	LAN, data-center, building (up to a few hundred meters)
10 Gb/s over 60 km	**Suitable**	Not suitable	Not suitable

Table 9: Comparison of ITU-T fiber categories

Characteristic	ITU-T G.652.D	ITU-T G.655	ITU-T G.657 (A-type)
Fiber type	Standard SMF	Non-zero dispersion-shifted SMF	Bend-insensitive SMF
Mode behaviour	Single-mode at 1550 nm	Single-mode at 1550 nm	Single-mode (A categories compatible with G.652.D)
Chromatic dispersion at 1550 nm	17 ps/(nm·km) → Δτ = 408 ps over 60 km	4.5 ps/(nm·km) → Δτ = 108 ps over 60 km	Same as G.652.D (compatible); value not given
Bending performance	Standard; entry loss 0.8 dB (given)	Standard; entry loss not specified	Excellent; entry loss 0.1 dB (given); radius ≈ 7.5–10 mm
Typical application	Access, metro and long-haul; most installed	DWDM long-haul and metro (less compensation)	FTTH, building entry, patch cords, tight routing
Long-distance suitability	Good with dispersion compensation	Very good	Good (used as a short section at the end)
10 Gb/s suitability at 60 km	Needs DCF (409 ps > 100 ps)	Still needs compensation (112 ps > 100 ps)	As G.652.D
Deployment considerations	Largest installed base, lowest cost, easy splicing	Newer/more costly; less common in access plants	A-types can be spliced to G.652.D with low loss

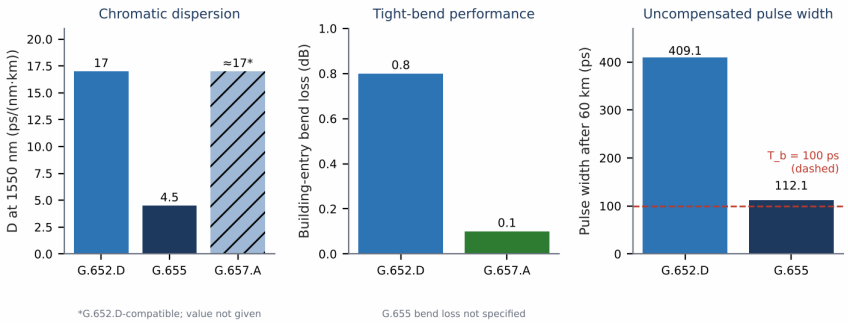


Figure 6: Comparison of G.652.D, G.655 and G.657.A using the values given in the assignment (chromatic dispersion, building-entry bend loss, uncompensated pulse width).

DESIGN DECISION 1

I recommend single-mode step-index propagation using ITU-T G.652.D fiber (Fiber A type), with G.657.A bend-insensitive fiber at the building entry, because

- V(A) = 1.951 < 2.405 and λc = 1257 nm < 1550 nm, so only LP_{01} propagates and **modal dispersion is eliminated**, whereas Fiber B has V = 21.30 and ≈ 227 modes (modal spread ≈ 3.03 µs over 60 km, i.e. 30 000 bit periods at 10 Gb/s);
- the cut-off value (1257 nm) is consistent with the G.652 cable cut-off requirement (≤ 1260 nm), so Fiber A corresponds to the G.652.D category;
- G.652.D is the most widely available standard fiber, and its chromatic dispersion (408 ps) can be compensated first-order with DCF (Stage 4);
- G.657.A is G.652.D-compatible and reduces the building-entry bend loss from 0.8 dB to 0.1 dB (Stage 3).

Viva tip

Remember the three numbers: V(A)=1.951 (single-mode), V(B)=21.30 (≈227 modes), λc(A)=1257 nm.

5. STAGE 3 – OPTICAL POWER BUDGET

A. Total attenuation coefficient

A.1 Combined attenuation coefficient

Given	α_abs = 0.04 dB/km, α_Ray = 0.11 dB/km, α_bend = 0.03 dB/km
Formula	α_total = α_abs + α_Ray + α_bend (losses in dB/km add)
Substitution	α_total = 0.04 + 0.11 + 0.03
Calculation	α_total = 0.18 dB/km
Answer	**α_total = 0.18 dB/km**
Engineering Interpretation	This single coefficient represents the combined effect of absorption, Rayleigh scattering and distributed bending; every kilometre of fiber costs 0.18 dB.

B. 60 km fiber loss

B.1 Fiber loss over 60 km

Given	α_total = 0.18 dB/km, L = 60 km
Formula	Loss_fiber = α_total × L
Substitution	Loss_fiber = 0.18 dB/km × 60 km
Calculation	Loss_fiber = 10.8 dB (Absorption 2.4 dB + Rayleigh 6.6 dB + Bending 1.8 dB)
Answer	**Fiber loss = 10.8 dB**
Engineering Interpretation	The fiber alone consumes 10.8 dB of the available budget. Starting from +3 dBm, the signal would already be at −7.8 dBm at the end of the fiber, before any connectors or splices.

C. Connector losses

C.1 Total connector loss

Given	4 connectors, 0.4 dB each
Formula	Loss_conn = N_c × α_c
Substitution	Loss_conn = 4 × 0.4 dB
Calculation	Loss_conn = 1.6 dB
Answer	**Connector loss = 1.6 dB**
Engineering Interpretation	Connectors are demountable and therefore lossy; they contribute 1.6 dB, more than any other passive element.

D. Splice losses

D.1 Total splice loss

Given	10 splices, 0.08 dB each
Formula	Loss_splice = N_s × α_s
Substitution	Loss_splice = 10 × 0.08 dB
Calculation	Loss_splice = 0.8 dB

Answer	**Splice loss = 0.8 dB**
Engineering Interpretation	Each fusion splice is five times better than a connector (0.08 vs 0.4 dB), but ten of them still add 0.8 dB.

E. Building-entry bend loss

Table 10: Building-entry bend loss for the two cases

Fiber at building entry	Bend loss (given)
G.652.D-type	0.8 dB
G.657.A-type	0.1 dB

The bend-insensitive G.657.A fiber reduces the tight-bend loss by **0.8 – 0.1 = 0.7 dB**, a reduction of 87.5 %.

F. Total link loss (without DCF)

F.1 Total link loss for both cases

Given	Fiber 10.8 dB, connectors 1.6 dB, splices 0.8 dB, entry bend 0.8 dB (G.652.D) or 0.1 dB (G.657.A)
Formula	Loss_total = Loss_fiber + Loss_conn + Loss_splice + Loss_entry
Substitution	G.652.D: 10.8 + 1.6 + 0.8 + 0.8 G.657.A: 10.8 + 1.6 + 0.8 + 0.1
Calculation	G.652.D: 14.0 dB G.657.A: 13.3 dB
Answer	**Total link loss = 14.0 dB (G.652.D entry) and 13.3 dB (G.657.A entry)**
Engineering Interpretation	The G.657.A case saves 0.7 dB of the budget at no cost to the main fiber.

Table 11: Power-budget comparison

Loss element	G.652.D entry (dB)	G.657.A entry (dB)
Absorption (0.04 × 60)	2.4	2.4
Rayleigh scattering (0.11 × 60)	6.6	6.6
Distributed bending (0.03 × 60)	1.8	1.8
Fiber subtotal	10.8	10.8
Connectors (4 × 0.4)	1.6	1.6
Splices (10 × 0.08)	0.8	0.8
Building-entry bend	0.8	0.1
Total link loss	**14.0**	**13.3**
Transmitter power	+3 dBm	+3 dBm
Received power (before DCF/amp)	**−11.0 dBm**	**−10.3 dBm**

G. Received optical power

G.1 Received power

Given	P_Tx = +3 dBm; Loss_total = 14.0 dB (G.652.D) / 13.3 dB (G.657.A)
Formula	P_Rx (dBm) = P_Tx (dBm) − Loss_total (dB)
Substitution	G.652.D: P_Rx = +3 dBm − 14.0 dB G.657.A: P_Rx = +3 dBm − 13.3 dB

Calculation	G.652.D: 3 − 14.0 = −11.0 G.657.A: 3 − 13.3 = −10.3
Answer	**P_Rx = −11.0 dBm (G.652.D), −10.3 dBm (G.657.A)**
Engineering Interpretation	In dBm arithmetic a loss in dB is simply subtracted from a power in dBm. Both values are well above the −20 dBm sensitivity. (Equivalent linear powers: −11.0 dBm ≈ 79 µW; −10.3 dBm ≈ 93 µW.)

H. System margin

H.1 Margin against sensitivity and the required 6 dB

Given	P_Rx = −11.0 / −10.3 dBm; sensitivity = −20 dBm; required margin = 6 dB
Formula	Margin = P_Rx − Sensitivity Surplus = Margin − Required margin
Substitution	G.652.D: Margin = −11.0 − (−20) G.657.A: Margin = −10.3 − (−20)
Calculation	G.652.D: 9.0 dB; surplus = 9.0 − 6 = 3.0 dB G.657.A: 9.7 dB; surplus = 9.7 − 6 = 3.7 dB
Answer	**Margin = 9.0 dB (G.652.D) and 9.7 dB (G.657.A); both exceed the required 6 dB**
Engineering Interpretation	The power budget alone is sufficient: the received power exceeds the minimum required level of −14 dBm by 3.0 dB and 3.7 dB. However, this margin does not yet include the DCF (Stage 4), which will change the conclusion.

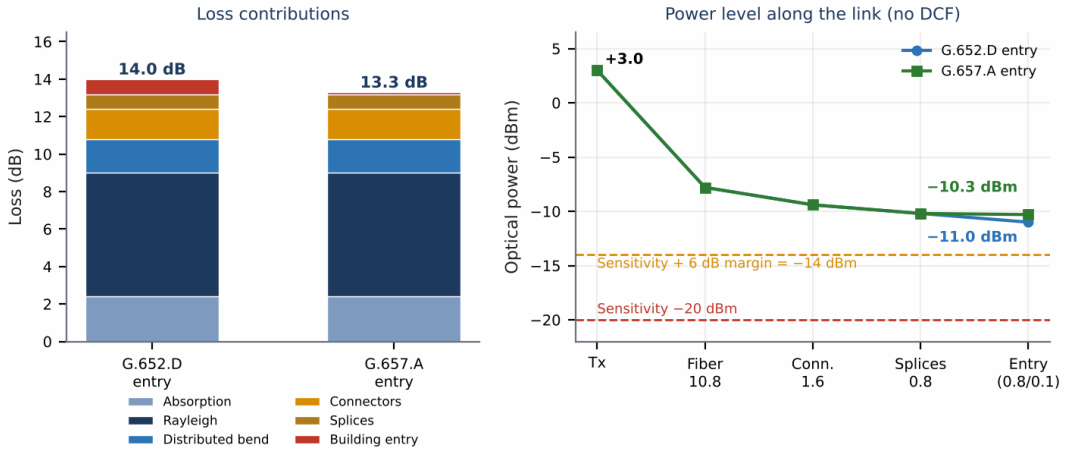


Figure 7: Optical power budget: loss contributions (left) and power level along the link without DCF (right).

I. Dominant loss mechanism

Table 12: Contribution of each specified fiber-loss mechanism

Mechanism	Loss over 60 km	Share of fiber loss	Share of total link loss (G.652.D / G.657.A)
Rayleigh scattering	6.6 dB	61.1 %	47.1 % / 49.6 %
Absorption	2.4 dB	22.2 %	17.1 % / 18.0 %
Distributed bending	1.8 dB	16.7 %	12.9 % / 13.5 %
Fiber total	10.8 dB	100 %	77.1 % / 81.2 %

Rayleigh scattering is the dominant specified fiber-loss mechanism (0.11 dB/km = 6.6 dB, 61 % of the fiber loss). It is, however, an intrinsic property of silica and cannot be reduced by installation practice. The **avoidable** bending-related losses (1.8 dB distributed + 0.8 dB building entry = 2.6 dB in the G.652.D case) and the 2.4 dB of

connectors and splices are where the designer can actually gain margin; therefore bending (distributed and at building entry) deserves the greatest optimization effort.

J. Loss-reduction methods

- **Use G.657.A bend-insensitive fiber at the building entry** and respect bend-radius limits in cable routing: saves 0.7 dB at the entry and reduces the distributed bending loss (up to 1.8 dB).
- **Reduce connector count and use clean, low-loss connectors** (cleaning, APC/PC polished ferrules, inspection): each connector removed saves 0.4 dB (e.g. replacing 2 connectors by fusion splices saves ≈ 0.64 dB).
- **Fusion-splice instead of mechanical splices and verify with an OTDR:** well-made fusion splices at or below 0.08 dB, re-doing poor splices, keeps the 0.8 dB splice allowance from growing.

DESIGN DECISION 2

I recommend the G.657.A-at-building-entry power-budget configuration because

• the power budget is **acceptable** without DCF: total loss 13.3 dB, received power −10.3 dBm, margin 9.7 dB ≥ 6 dB (G.652.D entry: 14.0 dB, −11.0 dBm, 9.0 dB);

• **G.657 should be used at the building entry**: it removes 0.7 dB of loss (0.8 → 0.1 dB) and raises the margin to 9.7 dB, which will later absorb part of the DCF loss;

• **Rayleigh scattering is the dominant specified loss (6.6 dB, 61 % of fiber loss)**, but it is intrinsic; the **bending-related loss** (1.8 dB + entry bend) and the **connectors/splices** (2.4 dB) require the greatest attention during optimization.

6. STAGE 4 – DISPERSION AND PMD MANAGEMENT

A. Modal dispersion

Single-mode fiber (Fiber A). All the optical energy travels in one mode (LP_{01}), so there is a single group velocity for a given wavelength and no mode-to-mode delay. Modal dispersion is therefore absent; only chromatic dispersion and PMD remain.

Multimode fiber (Fiber B). The ≈ 227 modes follow different path lengths. The slowest (steepest) ray travels farther than the axial ray, and the arrival-time spread for a step-index fiber is $\Delta\tau_modal \approx (n_1 L/c)\cdot(n_1 - n_2)/n_2$. **Graded-index fiber** has an index that decreases parabolically away from the axis: rays that travel farther spend more time in the lower-index (faster) outer region, so all modes arrive almost together. The spread falls to $\Delta\tau \approx n_1 L\Delta^2/(8c)$, with $\Delta = (n_1 - n_2)/n_1$.

A.1 Modal dispersion of Fiber B over 60 km (illustration)

Given	$n_1 = 1.480$, $n_2 = 1.465$, $L = 60$ km, $c = 3 \times 10^8$ m/s
Formula	Step-index: $\Delta\tau = (n_1 L/c)\cdot(n_1 - n_2)/n_2$ Graded-index: $\Delta\tau = n_1 L\Delta^2/(8c)$, $\Delta = (n_1 - n_2)/n_1$
Substitution	$n_1 L/c = 1.480 \times 60\,000 / (3 \times 10^8) = 2.96 \times 10^{-4}$ s Step: $2.96 \times 10^{-4} \times (0.015/1.465)$; Graded: $\Delta = 0.015/1.480 = 0.01014$; $\Delta^2 = 1.027 \times 10^{-4}$
Calculation	Step-index: $\Delta\tau = 2.96 \times 10^{-4} \times 0.01024 = 3.03 \times 10^{-6}$ s Graded-index: $\Delta\tau = 2.96 \times 10^{-4} \times 1.027 \times 10^{-4} / 8 = 3.80 \times 10^{-9}$ s
Answer	**Step-index $\Delta\tau \approx 3.03$ µs; graded-index $\Delta\tau \approx 3.80$ ns**
Engineering Interpretation	Even the graded-index spread (3.8 ns = 38 bit periods) is far above the 100 ps bit period, and the step-index spread (3.03 µs) is 30 000 bit periods. A rough maximum bit rate $1/(2\Delta\tau)$ is ≈ 165 kb/s (step) or ≈ 132 Mb/s (graded). Multimode fiber is therefore unsuitable for 10 Gb/s over 60 km. ($c = 3 \times 10^8$ m/s is a standard constant; it is not an assignment value.)

B. Chromatic dispersion

B.1 Chromatic broadening, G.652.D

Given	$D = 17$ ps/(nm·km), $\Delta\lambda = 0.4$ nm, $L = 60$ km
Formula	$\Delta\tau = D \times \Delta\lambda \times L$
Substitution	$\Delta\tau = 17$ ps/(nm·km) × 0.4 nm × 60 km
Calculation	$17 \times 0.4 = 6.8$ ps/km; $6.8 \times 60 = 408$ ps
Answer	**$\Delta\tau$(G.652.D) = 408 ps**
Engineering Interpretation	A source of spectral width 0.4 nm spreads each pulse by 408 ps over 60 km, which is four times the 100 ps bit period.

B.2 Chromatic broadening, G.655

Given	$D = 4.5$ ps/(nm·km), $\Delta\lambda = 0.4$ nm, $L = 60$ km
Formula	$\Delta\tau = D \times \Delta\lambda \times L$
Substitution	$\Delta\tau = 4.5$ ps/(nm·km) × 0.4 nm × 60 km
Calculation	$4.5 \times 0.4 = 1.8$ ps/km; $1.8 \times 60 = 108$ ps
Answer	**$\Delta\tau$(G.655) = 108 ps**

Engineering Interpretation	G.655 has 3.8 times less dispersion (17/4.5), but 108 ps still exceeds the 100 ps bit period.

C. Total pulse width (RSS)

C.1 Output pulse width

Given	τ_{input} = 30 ps; $\Delta\tau$ = 408 ps (G.652.D), 108 ps (G.655)
Formula	$\tau_{total} = \sqrt{(\tau_{input}^2 + \Delta\tau^2)}$
Substitution	G.652.D: $\tau_{total} = \sqrt{(30^2 + 408^2)} = \sqrt{(900 + 166\,464)}$ G.655: $\tau_{total} = \sqrt{(30^2 + 108^2)} = \sqrt{(900 + 11\,664)}$
Calculation	G.652.D: $\sqrt{167\,364}$ = 409.1 ps G.655: $\sqrt{12\,564}$ = 112.1 ps
Answer	**τ_{total} = 409.1 ps (G.652.D) and 112.1 ps (G.655)**
Engineering Interpretation	The RSS rule is used because the input width and the dispersion broadening are independent and add in quadrature. The input pulse (30 ps) is negligible compared with the broadening, so the output width is almost entirely dispersion.

D. Bit-period comparison

Table 13: Pulse width compared with the 10 Gb/s bit period

Fiber	τ_{total}	Bit period T_b	τ_{total} / T_b	Satisfactory ($\tau_{total} < T_b$)?
G.652.D	409.1 ps	100 ps	4.09	No — overlaps ≈ 4 bit slots
G.655	112.1 ps	100 ps	1.12	No — slightly more than one bit slot

- **Pulse overlap:** a pulse wider than the bit period spills into the adjacent time slots.
- **Inter-symbol interference (ISI):** energy from neighboring bits adds to each decision instant, closing the eye and raising BER; G.652.D (409 ps) would smear each bit over about four slots.
- **10 Gb/s suitability:** neither uncompensated option meets τ_{total} < 100 ps, so **dispersion compensation is required in both cases**, although G.655 needs far less.

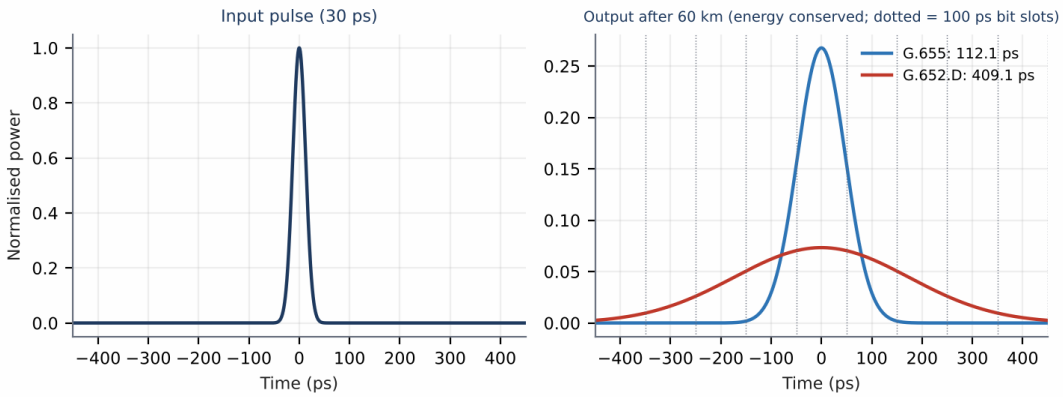


Figure 8: Chromatic dispersion and pulse broadening: 30 ps input pulse and the output after 60 km of G.655 (112.1 ps) and G.652.D (409.1 ps); dotted lines mark 100 ps bit slots.

E. DCF compensation

Condition for complete first-order compensation: the accumulated dispersion of the transmission fiber must be cancelled by the DCF: $\mathbf{D_1 \cdot L_1 + D_DCF \cdot L_DCF = 0}$, i.e. $L_DCF = D_1 \cdot L_1 / |D_DCF|$.

E.1 DCF length for the G.652.D link

Given	D_1 = 17 ps/(nm·km), L_1 = 60 km, D_DCF = −100 ps/(nm·km)

Formula	$D_1L_1 + D_DCF \cdot L_DCF = 0$ $L_DCF = D_1L_1 / \lvert D_DCF \rvert$
Substitution	$L_DCF = (17 \times 60) / 100$
Calculation	Accumulated dispersion = 17 × 60 = 1020 ps/nm L_DCF = 1020 / 100 = 10.2 km Check: 17×60 + (−100)×10.2 = 1020 − 1020 = 0 OK
Answer	**L_DCF = 10.2 km**
Engineering Interpretation	Only 10.2 km of DCF is needed to cancel 60 km of standard fiber because the DCF has a dispersion 5.9 times larger in magnitude and of opposite sign. Net first-order dispersion becomes zero, so the pulse returns to ≈ its original width.

E.2 Added DCF attenuation

Given	L_DCF = 10.2 km, α_DCF = 0.5 dB/km
Formula	Loss_DCF = α_DCF × L_DCF
Substitution	Loss_DCF = 0.5 dB/km × 10.2 km
Calculation	Loss_DCF = 5.1 dB
Answer	**DCF loss = 5.1 dB**
Engineering Interpretation	The compensating fiber is five times lossier than the main fiber per kilometre (0.5 vs 0.18 dB/km), so the compensation costs 5.1 dB of power budget.

E.3 New total loss, received power and margin (with DCF)

Given	Link loss 13.3 dB (G.657.A) or 14.0 dB (G.652.D); DCF loss 5.1 dB; P_Tx = +3 dBm; sensitivity −20 dBm; required margin 6 dB
Formula	Loss_new = Loss_total + Loss_DCF P_Rx = P_Tx − Loss_new; Margin = P_Rx − Sensitivity
Substitution	G.657.A: Loss_new = 13.3 + 5.1; P_Rx = 3 − 18.4; Margin = −15.4 − (−20) G.652.D: Loss_new = 14.0 + 5.1; P_Rx = 3 − 19.1; Margin = −16.1 − (−20)
Calculation	G.657.A: 18.4 dB; −15.4 dBm; 4.6 dB (shortfall 6 − 4.6 = 1.4 dB) G.652.D: 19.1 dB; −16.1 dBm; 3.9 dB (shortfall 6 − 3.9 = 2.1 dB)
Answer	**With DCF: P_Rx = −15.4 dBm (G.657.A) / −16.1 dBm (G.652.D); margin 4.6 dB / 3.9 dB — below the required 6 dB**
Engineering Interpretation	Solving the dispersion problem has used 5.1 dB of budget and the system margin has fallen **below** the required 6 dB. An optical amplifier is therefore needed. Section 7 shows the amplifier gives +14 dB, bringing the received power to −1.4 dBm (G.657.A) and −2.1 dBm (G.652.D), i.e. margins of **18.6 dB** and **17.9 dB**. (Only 1.4 dB of gain is required to restore the 6 dB margin; the receiver's maximum input power is not specified in the assignment, so it should be checked, and the gain adjusted or an attenuator used if necessary.)

Table 14: Effect of the DCF and amplifier on the power budget

Case	Total loss (dB)	P_Rx (dBm)	Margin vs −20 dBm (dB)	≥ 6 dB?
G.657.A, no DCF	13.3	−10.3	9.7	Yes
G.652.D, no DCF	14.0	−11.0	9.0	Yes
G.657.A + DCF (10.2 km)	18.4	−15.4	4.6	**No**
G.652.D + DCF (10.2 km)	19.1	−16.1	3.9	**No**

G.657.A + DCF + 14 dB amplifier	18.4 − 14 = 4.4 net	−1.4	18.6	Yes
G.652.D + DCF + 14 dB amplifier	19.1 − 14 = 5.1 net	−2.1	17.9	Yes

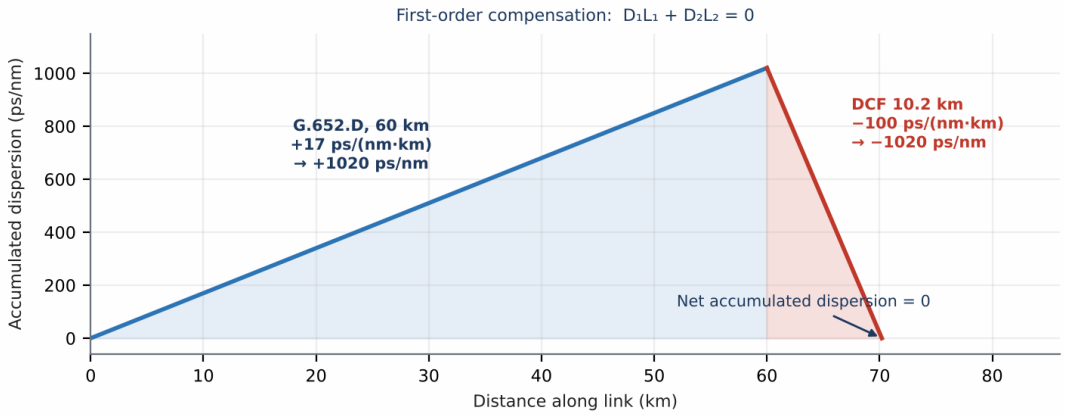


Figure 9: DCF compensation: accumulated dispersion rises to +1020 ps/nm over 60 km of G.652.D and is brought back to zero by 10.2 km of DCF.

F. PMD

F.1 Link PMD and comparison with the limits

Given	PMD coefficient = 0.1 ps/√km; L = 60 km; limit = 0.1·T_b; T_b = 100 ps (10 Gb/s) and ≈ 10 ps (100 Gb/s)
Formula	PMD = 0.1 × √L Limit = 0.1 × T_b
Substitution	PMD = 0.1 × √60
Calculation	√60 = 7.746; PMD = 0.7746 ps 10 Gb/s limit = 0.1 × 100 = 10 ps; 100 Gb/s limit = 0.1 × 10 = 1 ps
Answer	**PMD = 0.775 ps; limit 10 ps (10 Gb/s); limit 1 ps (100 Gb/s)**
Engineering Interpretation	At 10 Gb/s the PMD uses only 7.7 % of the allowed 10 ps. At 100 Gb/s it uses 77.5 % of the 1 ps limit (0.225 ps spare): it is still acceptable, but the margin is thin. The upgrade therefore needs a fiber measurement of the actual PMD (and any PMD from the DCF, which is not given) or a coherent receiver with electronic PMD compensation.

G. PMD mitigation

- **Use low-PMD fiber** (PMD coefficient ≤ 0.1 ps/√km, as assumed), well-manufactured cable and gentle installation (avoid twists, tight bends and crushing).
- **Measure the installed link** (PMD test) and reject sections with high local PMD; replace or re-route if needed.
- **Use PMD compensation for the 100 Gb/s upgrade:** optical PMD compensators or, preferably, coherent detection with digital signal processing.
- **Keep section length short or regenerate** if the limit would be exceeded (PMD grows with √L).

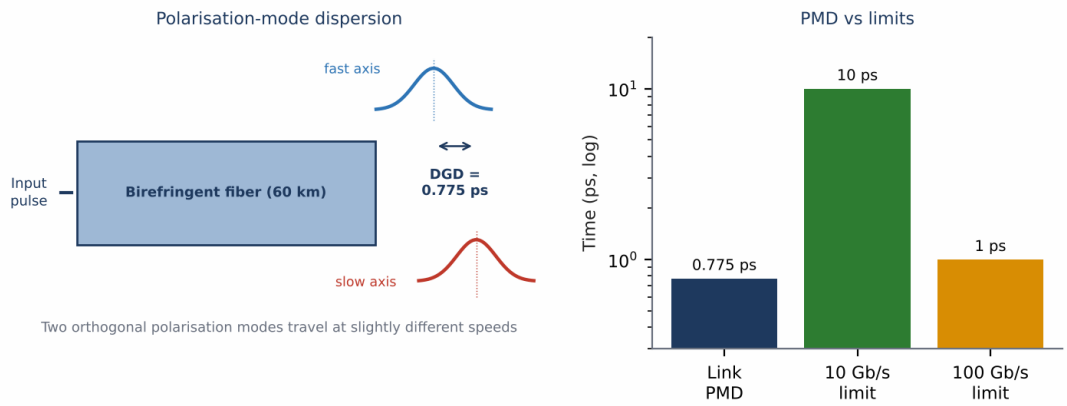


Figure 10: PMD / polarization illustration: differential group delay between two polarization modes (left) and link PMD versus the 10 Gb/s and 100 Gb/s limits (right).

H. Birefringence

Birefringence B = |n_x − n_y| is the refractive-index difference between the two orthogonal polarization axes. It arises from core non-circularity, stress and bending. A **polarization beat length** L_B = λ/B is the distance over which the two polarization components accumulate a 2π phase difference. A short beat length means strong birefringence, which keeps a launched polarization stable (PM fiber); a long beat length means weak birefringence and polarization that wanders easily.

H.1 Beat length of standard fiber

Given	$\lambda = 1550\ \text{nm} = 1.55 \times 10^{-6}\ \text{m}$; $B = 1 \times 10^{-7}$
Formula	$L_B = \lambda / B$
Substitution	$L_B = 1.55 \times 10^{-6}\ \text{m} / (1 \times 10^{-7})$
Calculation	$L_B = 15.5\ \text{m}$
Answer	**L_B (standard fiber) = 15.5 m**
Engineering Interpretation	Polarization in standard fiber changes slowly and randomly along its length; this weak and random birefringence is the origin of PMD.

H.2 Beat length of PM fiber

Given	$\lambda = 1.55 \times 10^{-6}\ \text{m}$; $B = 4 \times 10^{-4}$
Formula	$L_B = \lambda / B$
Substitution	$L_B = 1.55 \times 10^{-6}\ \text{m} / (4 \times 10^{-4})$
Calculation	$L_B = 3.875 \times 10^{-3}\ \text{m} = 3.875\ \text{mm}$
Answer	**L_B (PM fiber) = 3.875 mm**
Engineering Interpretation	The beat length is about 4000 times shorter, so polarization is locked to a fixed axis.

Is PM fiber required? No. PM fiber is used where a fixed polarization must be maintained over short lengths (interferometric sensors, modulators, coherent components). A 60 km telecom link has a PMD of only 0.775 ps (below both limits) with ordinary low-PMD G.652.D fiber, and the receiver is polarization-insensitive for intensity modulation. PM fiber would be expensive, would be lossier and harder to splice, and is not part of the G.652/G.655/G.657 standards. It is therefore **not needed**.

DESIGN DECISION 3

I recommend a G.652.D (low-PMD) fiber with a 10.2 km DCF module and standard (non-PM) fiber dispersion/PMD configuration because

- uncompensated dispersion is unacceptable: τ_total = 409.1 ps (4.09 T_b) for G.652.D and 112.1 ps (1.12 T_b) even for G.655;
- 10.2 km of DCF (D·L = −1020 ps/nm) cancels the 1020 ps/nm of 60 km G.652.D completely to first order, at a cost of 5.1 dB which the amplifier (+14 dB) more than recovers (P_Rx −1.4 dBm, margin 18.6 dB);
- PMD = 0.775 ps 10 ps (10 Gb/s) and < 1 ps (100 Gb/s), so standard low-PMD fiber is sufficient and **PM fiber is not required** (L_B = 15.5 m vs 3.875 mm); PMD is managed by low-PMD fiber, measurement and, for 100 Gb/s, coherent/electronic compensation.
- G.655 would need only 2.7 km of DCF (1.35 dB), see Section 10 — a valid alternative, but G.652.D with G.657.A is chosen because it matches the installed base and the supplied G.652.D/G.657.A building-entry data.

7. STAGE 5 – OSNR AND AMPLIFICATION

A. OSNR concept

OSNR (optical signal-to-noise ratio) compares the signal power with the amplified-spontaneous-emission (ASE) noise power from the optical amplifier. It is the correct quality metric for an amplified link because an amplifier increases signal and noise together; what matters for bit errors is how far the signal stands above the noise. By convention the noise is measured in a **0.1 nm reference bandwidth** (≈ 12.5 GHz at 1550 nm), so that OSNR values from different systems can be compared. When both quantities are in dBm, OSNR (dB) = P_signal (dBm) − P_noise (dBm).

B. Initial OSNR

B.1 Initial OSNR

Given	P_signal = −16 dBm; P_ASE = −39 dBm (0.1 nm reference bandwidth)
Formula	OSNR (dB) = P_signal (dBm) − P_noise (dBm)
Substitution	OSNR = (−16) − (−39)
Calculation	OSNR = −16 + 39 = 23 dB
Answer	**Initial OSNR = 23 dB**
Engineering Interpretation	The signal is 23 dB (a factor of ≈ 200) above the ASE noise: comfortably above the required 17 dB. The given signal power (−16 dBm) corresponds closely to the pre-amplifier power of the G.652.D + DCF design (−16.1 dBm, Section 6.E), so the OSNR study is a conservative check of the compensated link.

C. Amplifier effect

C.1 Signal level after the amplifier

Given	P_signal,in = −16 dBm; gain = 14 dB
Formula	P_signal,out = P_signal,in + G
Substitution	P_signal,out = −16 dBm + 14 dB
Calculation	P_signal,out = −2 dBm
Answer	**Signal = −2 dBm**
Engineering Interpretation	The amplifier raises the signal by 14 dB (a factor of 25).

C.2 ASE-noise level after the amplifier

Given	P_ASE,in = −39 dBm; noise increase = 16 dB
Formula	P_ASE,out = P_ASE,in + ΔN
Substitution	P_ASE,out = −39 dBm + 16 dB

Calculation	P_ASE,out = −23 dBm
Answer	**ASE noise = −23 dBm**
Engineering Interpretation	The noise rises by more than the signal (16 dB against 14 dB) because the amplifier adds its own ASE noise.

D. Final OSNR

D.1 OSNR after amplification

Given	P_signal,out = −2 dBm; P_ASE,out = −23 dBm
Formula	OSNR = P_signal − P_noise
Substitution	OSNR = (−2) − (−23)
Calculation	OSNR = −2 + 23 = 21 dB (equivalently 23 dB + 14 dB − 16 dB = 21 dB)
Answer	**Final OSNR = 21 dB**
Engineering Interpretation	The amplifier degrades OSNR by 2 dB (the noise gain exceeds the signal gain by 2 dB) while raising the absolute power by 14 dB.

E. OSNR comparison

Table 15: OSNR comparison

Quantity	Value
Initial OSNR	23 dB
Final OSNR (after amplifier)	21 dB
Change due to amplifier	−2 dB
Required OSNR	17 dB
Remaining OSNR margin = 21 − 17	**4 dB**

The final OSNR (21 dB) exceeds the requirement (17 dB) by **4 dB**, so the amplifier is acceptable from the OSNR point of view, but with a modest margin that should be protected.

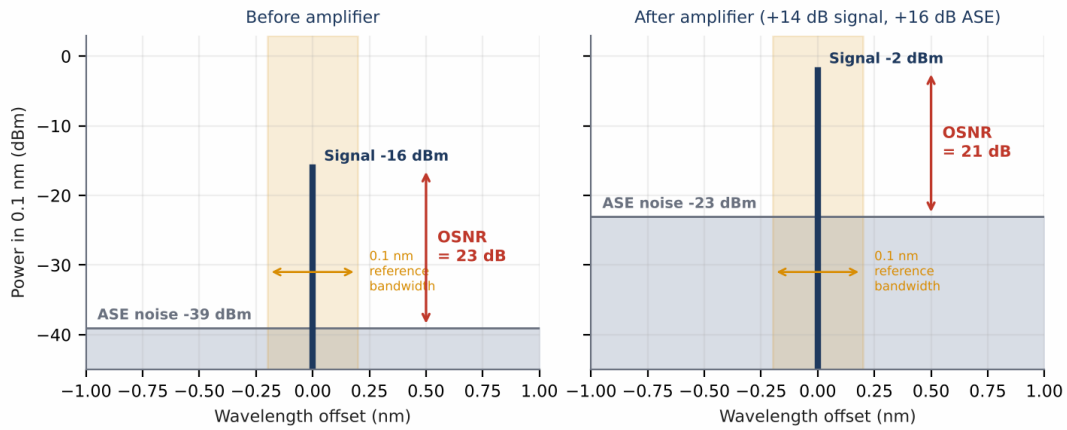


Figure 11: OSNR and ASE-noise illustration: spectrum before (OSNR 23 dB) and after (OSNR 21 dB) the optical amplifier, with the 0.1 nm reference bandwidth.

F. Effect of poor OSNR

- **Signal quality:** noise blurs the "1" and "0" levels, closing the eye diagram.
- **Q-factor:** $Q = (\mu_1 - \mu_0)/(\sigma_1 + \sigma_0)$; more noise gives larger σ and lower Q.
- **BER:** $BER \approx \frac{1}{2}\cdot\mathrm{erfc}(Q/\sqrt{2})$; for example Q = 6 gives BER ≈ 10^{-9}. A drop in OSNR therefore raises BER rapidly.

- **System margin:** the OSNR deficit must be paid for with more received power or a shorter link, consuming the 6 dB margin; below the required 17 dB the link would no longer meet its BER target even though the received power looks adequate.

G. OSNR improvement

- **Increase the signal level entering the amplifier / place the amplifier early (pre-amplifier or booster at the right point):** a higher input power gives a higher OSNR for the same ASE; avoid amplifying a very weak signal (e.g. put gain before the lossy DCF or split the DCF loss around the amplifier).
- **Use a low-noise-figure amplifier (e.g. EDFA) with a suitable gain setting and an optical filter** to limit out-of-band ASE; also minimize loss ahead of the amplifier (low-loss connectors, shorter DCF) and avoid unnecessary cascaded amplifiers.

DESIGN DECISION 4
I recommend a single optical-amplifier (+14 dB, 16 dB ASE increase) amplification/OSNR strategy because
• initial OSNR = 23 dB, final OSNR = 21 dB ≥ 17 dB required, leaving **4 dB OSNR margin**;
• the amplifier restores the power lost in the DCF: received power rises from −15.4 dBm to −1.4 dBm and the system margin from 4.6 dB (< 6 dB, not acceptable) to 18.6 dB;
• the 2 dB OSNR penalty is small and acceptable for one amplifier in a 60 km link; the strategy should be accepted for 10 Gb/s, with the OSNR margin re-checked for 100 Gb/s.

8. STAGE 6 – COMPLETE LINK VERIFICATION

The table below collects the results of Stages 2–5. All values are taken from the calculations above (final design: G.652.D 60 km, G.657.A at the building entry, 10.2 km DCF, 14 dB amplifier).

Table 16: Final verification table

Requirement	Target	Calculated / achieved	Satisfactory?	Engineering comment
Modal condition / fiber category	Single-mode, standard ITU-T fiber	V = 1.951 < 2.405; λc = 1257 nm; G.652.D (+ G.657.A entry)	Yes	Only LP_{01} propagates; no modal dispersion. Fiber B rejected (V = 21.30, ≈ 227 modes).
Fiber loss / link loss	Within budget	Fiber 10.8 dB (0.18 dB/km); link 13.3 dB (G.657.A), 14.0 dB (G.652.D)	Yes	Rayleigh dominant (6.6 dB); bending/connectors/splices are the avoidable losses.
Received optical power	≥ −14 dBm (−20 dBm + 6 dB)	−10.3 dBm (no DCF); −15.4 dBm (with DCF); −1.4 dBm (with DCF + amplifier)	Yes (with amp.)	Without the amplifier the DCF leaves only 4.6 dB margin (< 6 dB).
System margin	≥ 6 dB	9.7 dB (no DCF); 4.6 dB (with DCF); 18.6 dB (with amp.)	Yes (with amp.)	Amplifier restores 14 dB; receiver overload limit should be checked.
Pulse broadening	τ_total < T_b = 100 ps	409.1 ps (G.652.D); 112.1 ps (G.655); ≈ 30 ps after first-order DCF compensation	No (uncompensated) → Yes (with DCF)	Uncompensated pulses overlap ≈ 4 bit slots; compensation required.
DCF requirement	$D_1L_1 + D_DCF \cdot L_DCF = 0$	L_DCF = 10.2 km; loss 5.1 dB	Yes	Net first-order dispersion 0 ps/nm; loss included in the budget.
PMD	≤ 0.1·T_b (10 ps at 10 Gb/s; 1 ps at 100 Gb/s)	0.775 ps	Yes (10 Gb/s); Yes, marginal (100 Gb/s)	7.7 % of the 10G limit; 77.5 % of the 100G limit.
Polarization performance	Acceptable without PM fiber	L_B = 15.5 m (standard) vs 3.875 mm (PM); PMD 0.775 ps	Yes	PM fiber not required for a transmission link.
OSNR	≥ 17 dB	Initial 23 dB; final 21 dB	Yes	4 dB margin; amplifier degrades OSNR by 2 dB.

Consistency with the original requirement. Every item of the original communication requirement is met by the final configuration for 10 Gb/s: the modal condition (single-mode, standard fiber), the loss budget (margin 18.6 dB after the DCF and amplifier), pulse integrity (first-order compensated), PMD (0.775 ps 10 ps), and OSNR (21 dB ≥ 17 dB). The only items that need attention for the 100 Gb/s upgrade are PMD (77.5 % of the limit), the very tight chromatic-dispersion tolerance of a 10 ps bit and the OSNR margin of 4 dB.

Note on the OSNR value. The OSNR study of the assignment is evaluated at the given −16 dBm signal and −39 dBm noise. This equals the pre-amplifier level of the G.652.D + DCF case (−16.1 dBm); the final G.657.A design has a pre-amplifier level of −15.4 dBm, i.e. 0.7 dB higher, so the 21 dB result is slightly conservative.

9. FINAL SYSTEM DESIGN

The final configuration is created from the results of Stages 1–6. The assignment's proposed sequence (transmitter → source → 60 km fiber → dispersion compensation → amplifier → receiver) is checked against the calculated power levels. The amplifier placed **after** the DCF recovers the power lost in both the fiber and the DCF, and brings the signal to a comfortable level (−1.4 dBm) at the receiver; because the −15.4 dBm pre-amplifier level is still 4.6 dB above the receiver sensitivity, the signal does not fall below the noise floor of the amplifier. The sequence is therefore **kept**. The G.657.A bend-insensitive section forms the final fiber section at the building entry, and the DCF module is placed at the receiving end (post-compensation).

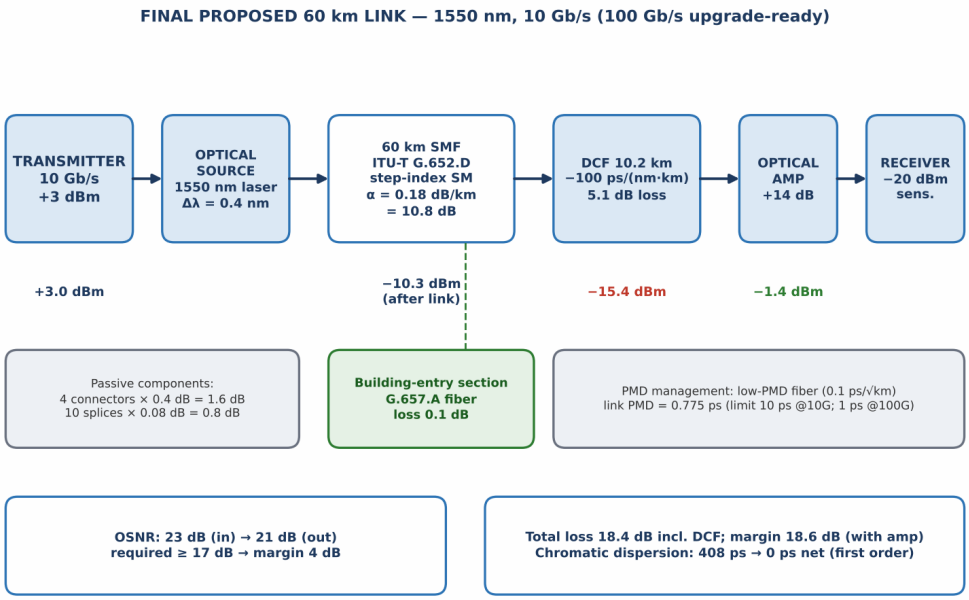


Figure 12: Final proposed 60 km system architecture with fiber type, ITU-T category, DCF length, amplifier gain, OSNR, PMD management, building-entry section and receiver.

Table 17: Final design parameter table

Design parameter	Adopted value
Fiber category	ITU-T G.652.D (60 km transmission fiber); G.657.A at the building entry
Propagation type	Single-mode, step-index (V = 1.951; λc = 1257 nm)
Wavelength	1550 nm
Data rate	10 Gb/s (T_b = 100 ps); 100 Gb/s upgrade provision
Distance	60 km
Fiber attenuation	0.18 dB/km → 10.8 dB (60 km)
Connectors / splices	4 × 0.4 dB = 1.6 dB; 10 × 0.08 dB = 0.8 dB
DCF length	10.2 km (D_DCF = −100 ps/(nm·km))
DCF attenuation	5.1 dB (0.5 dB/km)
Total loss incl. DCF	18.4 dB (G.657.A entry)
Amplifier gain	+14 dB (ASE noise +16 dB)
OSNR	23 dB → 21 dB (required 17 dB; margin 4 dB)
PMD value	0.775 ps (limit 10 ps at 10 Gb/s, 1 ps at 100 Gb/s); PM fiber not required

Received power	−15.4 dBm before the amplifier; −1.4 dBm after the +14 dB amplifier
System margin	4.6 dB before the amplifier; 18.6 dB with the amplifier (required ≥ 6 dB)
Building-entry fiber choice	G.657.A (0.1 dB bend loss instead of 0.8 dB)

How the design choices work together. (1) Single-mode G.652.D removes modal dispersion. (2) The power budget (13.3 dB) is acceptable alone, and the G.657.A entry saves 0.7 dB. (3) The 10.2 km DCF removes the 408 ps of chromatic dispersion but costs 5.1 dB, which would leave only 4.6 dB margin. (4) The +14 dB amplifier recovers this, producing 18.6 dB margin, at an OSNR of 21 dB (≥ 17 dB). (5) PMD (0.775 ps) is within both limits, so ordinary low-PMD fiber is sufficient. Each stage corrects the weakness left by the previous one; together they satisfy the requirement.

10. ENGINEERING DESIGN DECISION

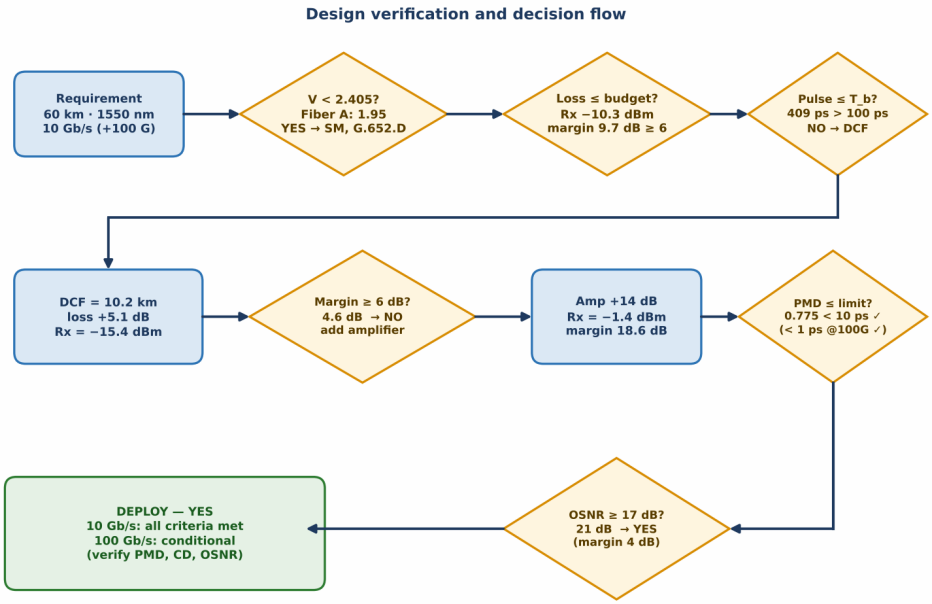


Figure 13: Final verification and design-decision flowchart.

WOULD YOU DEPLOY THE DESIGN?

YES — deploy for the 10 Gb/s service, with a conditional upgrade path to 100 Gb/s.

Table 18: Evidence supporting the deployment decision

Criterion	Evidence	Judgement
Fiber / modal suitability	V = 1.951 < 2.405, λc = 1257 nm: single-mode; no modal dispersion	Pass
ITU-T category	G.652.D transmission fiber, G.657.A at building entry (spliceable)	Pass
Propagation performance	Single-mode: bandwidth limited only by chromatic dispersion and PMD, both compensated/managed	Pass
Power budget	Total loss 13.3 dB (18.4 dB with DCF)	Pass
Received power	−10.3 dBm (no DCF); −15.4 dBm (with DCF); −1.4 dBm (with amplifier)	Pass
System margin	18.6 dB ≥ 6 dB with the amplifier; 4.6 dB without it (fails)	Pass — amplifier is mandatory
Dominant loss	Rayleigh 6.6 dB (intrinsic); avoidable: bend and entry bend 2.6 dB, connectors/splices 2.4 dB — controlled by G.657.A and good installation	Pass
Chromatic dispersion	408 ps (τ_total 409.1 ps) = 4.09 T_b before compensation	Needs DCF

DCF	10.2 km compensates to zero first-order; costs 5.1 dB	Pass
PMD	0.775 ps ≤ 10 ps (10 Gb/s); 0.775 ps ≤ 1 ps (100 Gb/s)	Pass (100G marginal)
OSNR and amplification	23 → 21 dB ≥ 17 dB; 4 dB margin	Pass
10 Gb/s operation	All criteria satisfied	Pass
100 Gb/s upgrade	Bit period 10 ps; PMD at 77.5 % of 1 ps limit; chromatic-dispersion tolerance far smaller; higher OSNR needed	Conditional

Conditions and redesign options

The answer is YES for 10 Gb/s because every requirement is met numerically. The 100 Gb/s upgrade is possible but conditional: PMD would be at 77.5 % of its 1 ps limit, the 10 ps bit period is very sensitive to residual dispersion (the first-order DCF cancels D but not slope or temperature drift), and the OSNR requirement for 100 Gb/s transmission is higher than for 10 Gb/s. Before upgrading, the installed link PMD and residual dispersion should be measured, and coherent detection with digital signal processing should be used.

Because the conclusion is "YES", no redesign is mandatory. A **recommended optimization** is to evaluate G.655 transmission fiber, because it also meets the requirement with a lower compensation penalty:

Table 19: Alternative evaluated: G.655 with a shorter DCF

Parameter	G.652.D + DCF (adopted)	G.655 + DCF (option)
Chromatic broadening Δτ	408 ps	108 ps
DCF length = D·L/|D_DCF|	10.2 km	2.7 km (4.5 × 60/100)
DCF loss	5.1 dB	1.35 dB
Total loss (with G.657.A entry, 0.1 dB)	18.4 dB	14.65 dB (13.3 + 1.35)
Received power before amplifier	−15.4 dBm	−11.65 dBm
Margin before amplifier	4.6 dB (< 6 dB)	8.35 dB (≥ 6 dB)
Comment	Needs the amplifier	Meets 6 dB margin even without the amplifier

G.655 is not chosen as the base design because (i) the assignment supplies the building-entry data only for G.652.D/G.657.A types, (ii) G.652.D has the largest installed base and is directly spliceable to G.657.A, and (iii) the amplifier already restores the margin. It is, however, a valid redesign if a future link must avoid the amplifier or carry multiple wavelengths.



Figure 15. Illustrative telecom optical-distribution rack; generated visual for deployment context, not site-specific evidence.

> **Viva tip**
> If asked "why use an amplifier if the margin was fine before DCF?" — because the 10.2 km DCF costs 5.1 dB and the margin drops to 4.6 dB, below the 6 dB requirement; the amplifier fixes this while OSNR stays at 21 dB.

11. FINAL CONCLUSION

DESIGN RESULTS AT A GLANCE

1.951 V-number · single-mode	**13.3 dB** Total link loss	**10.2 km** DCF length (5.1 dB)
18.6 dB System margin (≥ 6 dB)	**0.775 ps** PMD (limit 10 ps)	**21 dB** Final OSNR (≥ 17 dB)

A 60 km, 1550 nm, 10 Gb/s link was designed from the supplied requirement. Fiber A is single-mode (V = 1.951; cut-off 1257 nm) and corresponds to ITU-T G.652.D, whereas Fiber B (V = 21.30, ≈ 227 modes) is rejected because of modal dispersion. The fiber attenuation is 0.18 dB/km (10.8 dB over 60 km); with connectors, splices and a G.657.A building entry the total loss is 13.3 dB, giving −10.3 dBm received power and a 9.7 dB margin. Chromatic dispersion broadens the pulse to 409.1 ps (G.652.D) or 112.1 ps (G.655) against the 100 ps bit period, so 10.2 km of DCF is required for G.652.D; it adds 5.1 dB of loss, reducing the margin to 4.6 dB, so a 14 dB amplifier is included, giving −1.4 dBm at the receiver and 18.6 dB margin. PMD is 0.775 ps (limits 10 ps and 1 ps), so PM fiber is not needed. OSNR falls from 23 dB to 21 dB, still 4 dB above the required 17 dB.

The configuration is suitable for deployment at 10 Gb/s. The 100 Gb/s upgrade is feasible but depends on measured PMD (77.5 % of its limit), tight residual-dispersion control and adequate OSNR.

12. REFERENCES

[1] ITU-T Recommendation G.652, “Characteristics of a single-mode optical fibre and cable,” International Telecommunication Union, Geneva, Switzerland, Nov. 2016.

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[7] B. E. A. Saleh and M. C. Teich, Fundamentals of Photonics, 2nd ed. Hoboken, NJ, USA: Wiley, 2007.

Note: All numerical input values are taken from the problem data in the assignment PDF. Standards are revised periodically, so the latest edition of each recommendation applies in practice.

Visual note: The cover artwork and Figures 14–15 are AI-generated illustrative visuals. They are included to explain the deployment context and do not depict the actual proposed route, equipment, or installation. The technical block diagrams, plots, and calculations are schematic/coursework illustrations.